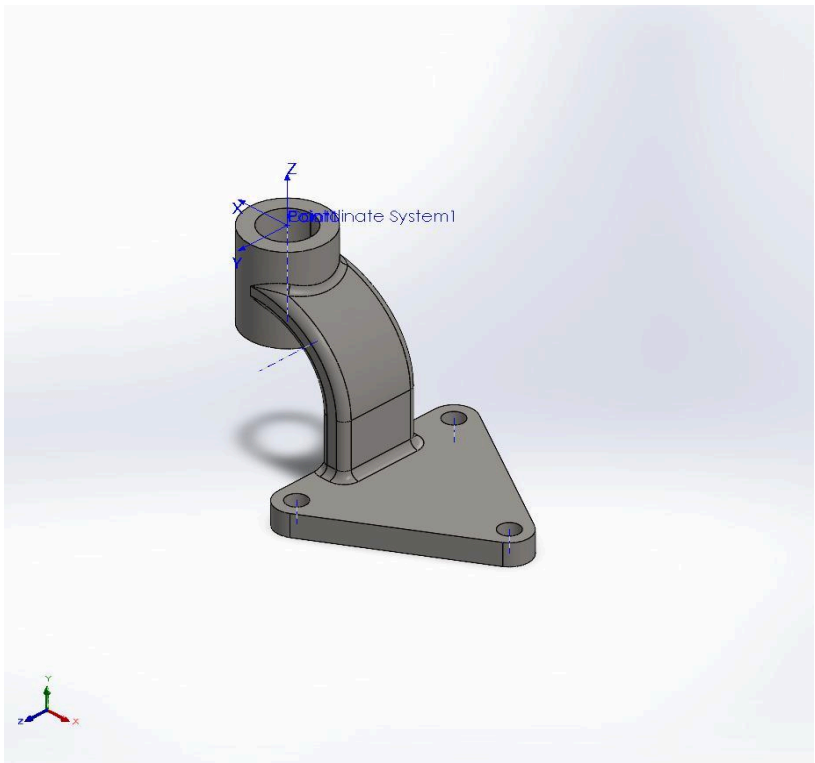


Mechanical Drive Pulley Support



Description

Static structural finite element analysis of a cast alloy steel pulley support bracket evaluating stress concentrations, structural deflections, and safety factors under combined horizontal bearing and vertical static loads. The study verifies mechanical integrity and identifies non-critical material volume for potential light-weighting and manufacturing optimization."

Simulation of Pulley Support

Date: Thursday, December 11, 2025
Designer: LUIS RAMIREZ
Study name: PULLEY SUPPORT STATIC ANALYSIS
Analysis type: Static

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Mechanical Drive Pulley Support

Assumptions

- **Linear Elastic Isotropic Material:** The pulley bracket material is assumed to be homogeneous, continuous, and isotropic throughout its volume. The material adheres to Hooke's Law, meaning all deformations stay within the elastic range with no permanent yield or plastic deformation.
- **Small Displacement and Small Strain Formulation:** Structural deflections are assumed to be very small compared to the overall size of the bracket (maximum displacement is 0.0093 inches). The software solves the equations on the undeformed part shape, ignoring large geometric non-linear effects.
- **Rigid Boundary Constraints:** The fixed boundary condition on the three base bolt holes assumes completely rigid fasteners with zero slip, zero bolt clearance, and no prying against the mounting surface.
- **Continuous Monolithic Body (Interaction Information):** Because this part is modeled and analyzed as a single solid body, there are no internal contact sets or joint interactions between parts (shown as No Data in the report). The bracket behaves as one solid piece of material from the base up through the collar.
- **Idealized Bearing Load:** The bearing load on the inner split face of the cylinder assumes smooth half-circle contact with a standard parabolic distribution. Pin rotation, friction, heating, and shaft misalignment are not included.
- **Static Loading and Room Temperature:** The loads (100 lbf bearing force and 20 lbf vertical force) are steady static forces applied at room temperature (298 Kelvin), meaning vibration, dynamic fatigue, and thermal stress are neglected.



Mechanical Drive Pulley Support

Model Information

Component Structure and Role: The component is a structural pulley support bracket designed to react belt and shaft loads. It includes three main functional sections:

- An upper cylindrical collar to support a shaft and bearing assembly.
- A curved support column that transfers loads and bending moments down to the base.
- A flat triangular base plate with three mounting holes for bolt anchoring.

Material Properties (Cast Alloy Steel):

- **Material Name:** Cast Alloy Steel (Linear Elastic Isotropic)
- **Elastic Modulus:** 27,557,000 psi (1.90×10^{11} N/m²)
- **Poisson's Ratio:** 0.26
- **Mass Density:** 0.264 lb/in³ (7,300 kg/m³)
- **Yield Strength:** 34,994 psi (2.41×10^8 N/m²)
- **Tensile Strength:** 64,989 psi (4.48×10^8 N/m²)
- **Part Mass and Weight:** 0.624 lb (0.283 kg) / 2.77 N

Fixtures and Restraints:

- **Fixed-1:** 3 cylindrical bolt hole surfaces on the base plate fixed in all directions.

Applied External Loads:

- **BearingLoads-1:** 100 lbf applied in the positive X-direction along the inner split face of the upper cylinder using Coordinate System 1.
- **Force-1:** 20 lbf applied downwards normal to the top flat face of the cylinder.

Mesh and Local Refinement:

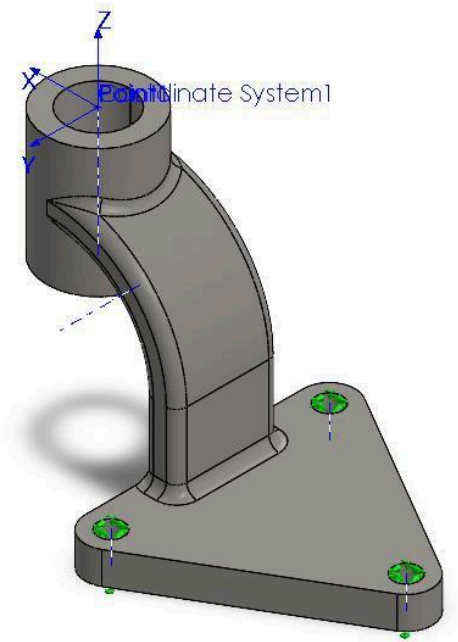
- **Mesh Type:** High Quality solid mesh using the blended curvature-based mesher with 16 Jacobian points.
- **Element Count:** 50,260 nodes and 32,645 elements with a maximum aspect ratio of 3.975 (99.8% of elements have an aspect ratio under 3, with 0% distorted elements).
- **Mesh Control-1:** Local mesh refinement applied to the 7 fillet faces connecting the curved column to the base plate using a 0.033 inch element size to capture high stress gradients accurately.



Mechanical Drive Pulley Support



Mechanical Drive Pulley Support



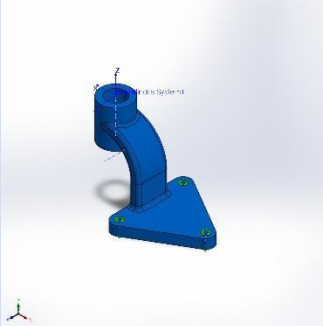
Model name: Pulley Support
Current Configuration: Default

Solid Bodies

Document Name and Reference	Treated As	Volumetric Properties	Document Path/Date Modified
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Mechanical Drive Pulley Support

Split Line1 	Solid Body	Mass:0.283141 kg Volume:3.87864e-05 m^3 Density:7,300 kg/m^3 Weight:2.77478 N	Support.SLDPRT Dec 10 22:03:49 2025
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Study Properties

Study name	PULLEY SUPPORT STATIC ANALYSIS
Analysis type	Static
Mesh type	Solid Mesh
Thermal Effect:	On
Thermal option	Include temperature loads
Zero strain temperature	298 Kelvin
Include fluid pressure effects from SOLIDWORKS Flow Simulation	Off
Solver type	Automatic
Inplane Effect:	Off
Soft Spring:	Off
Inertial Relief:	Off
Incompatible bonding options	Automatic
Contact penalty stiffness scale factor	1
Large displacement	Off
Compute free body forces	On
Friction	Off
Use Adaptive Method:	Off



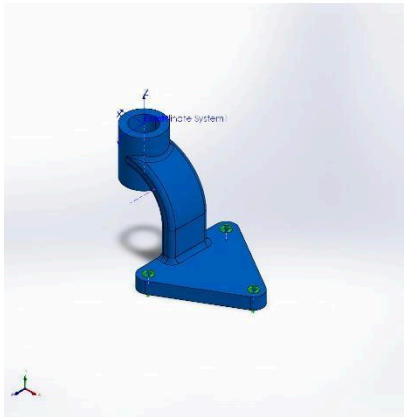
Mechanical Drive Pulley Support

Result folder	SOLIDWORKS document support)
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Units

Unit system:	SI (MKS)
Length/Displacement	mm
Temperature	Kelvin
Angular velocity	Rad/sec
Pressure/Stress	N/m ²

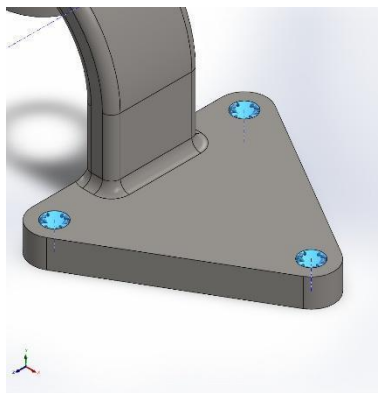
Material Properties

Model Reference	Properties	Components
	Name: Cast Alloy Steel Model type: Linear Elastic Isotropic Default failure criterion: Unknown Yield strength: 2.41275e+08 N/m² Tensile strength: 4.48082e+08 N/m² Elastic modulus: 1.9e+11 N/m² Poisson's ratio: 0.26 Mass density: 7,300 kg/m³ Shear modulus: 7.8e+10 N/m² Thermal expansion coefficient: 1.5e-05 /Kelvin	SolidBody 1(Split Line1)(Pulley Support)
Curve Data: N/A		



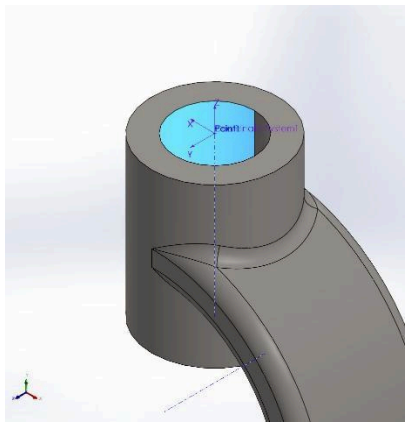
Mechanical Drive Pulley Support

Loads and Fixtures

Fixture name	Fixture Image	Fixture Details
Fixed-1		Entities: 3 face(s) Type: Fixed Geometry

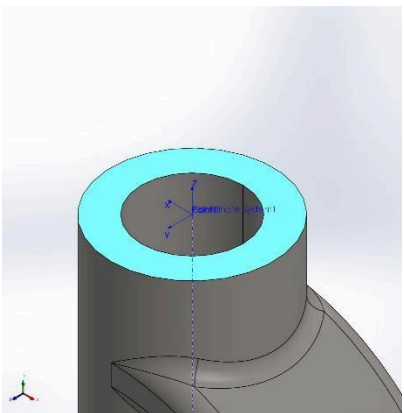
Resultant Forces

Components	X	Y	Z	Resultant
Reaction force(N)	444.799	88.958	0.00399351	453.608
Reaction Moment(N.m)	0	0	0	0

Load name	Load Image	Load Details
BearingLoads-1		Entities: 1 face(s) Coordinate Coordinate System: System1 Force Values: 100 0 0 lbf



Mechanical Drive Pulley Support

Force-1		Entities: 1 face(s) Type: Apply normal force Value: 20 lbf
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Interaction Information

Component Nature: The pulley support is modeled and analyzed as a single, continuous solid body rather than an assembly of separate components.

Contact Sets and Interfaces: Because there are no mating surfaces, sliding contacts, or bonded joints between separate solid bodies, no contact sets or interaction pairs were defined in this study.

Report Designation ("No Data"): The simulation software automatically displays "No Data" for this section because global contact conditions and localized contact formulations (such as bonded, no penetration, or shrink fit) apply strictly to multi-body parts or assemblies.

Structural Continuity: The part maintains 100% material continuity throughout, with standard seamless load transfer from the upper bearing collar, through the curved support arm, and into the triangular base plate.

Mesh information

Mesh type	Solid Mesh
Mesher Used:	Blended curvature-based mesh
Jacobian points for High quality mesh	16 Points



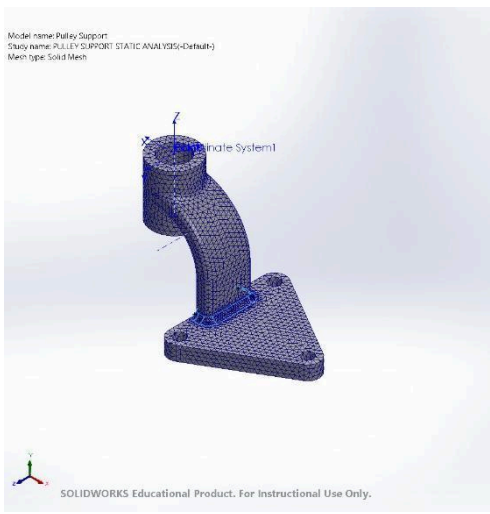
Mechanical Drive Pulley Support

Maximum element size	0.0966517 in
Minimum element size	0.0322169 in
Mesh Quality	High

Mesh information - Details

Total Nodes	50260
Total Elements	32645
Maximum Aspect Ratio	3.975
% of elements with Aspect Ratio < 3	99.8
Percentage of elements with Aspect Ratio > 10	0
Percentage of distorted elements	0
Time to complete mesh(hh:mm:ss):	00:00:03
Computer name:	ANT

Mesh Control Information:

Mesh Control Name	Mesh Control Image	Mesh Control Details
Control-1	 Model name: Pulley Support Study name: PULLEY SUPPORT STATIC ANALYSIS (Default) Mesh type: Solid Mesh Coordinate System1 SOLIDWORKS Educational Product. For Instructional Use Only.	Entities: 7 face(s) Units: in Size: 0.0333283 Ratio: 0.0333283



Mechanical Drive Pulley Support

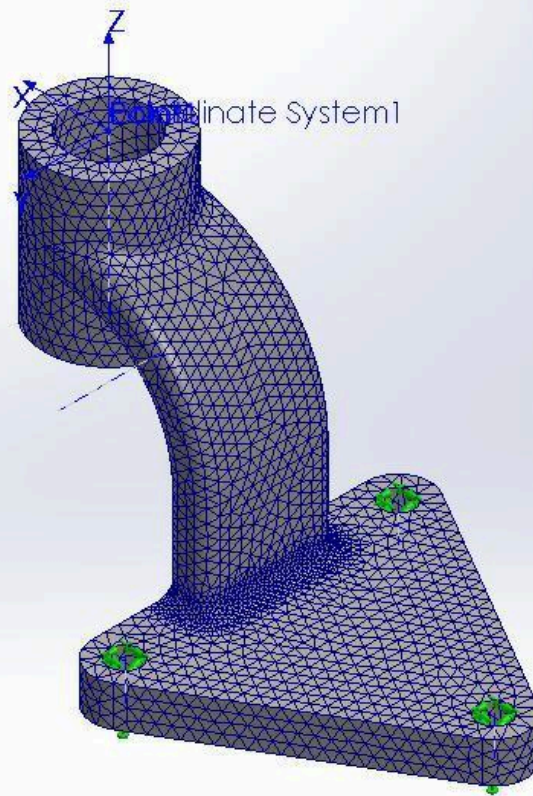
Mesh Quality Plots

Name	Type	Min	Max
Quality1	Mesh	-	-



Mechanical Drive Pulley Support

Model name: Pulley Support
Study name: PULLEY SUPPORT STATIC ANALYSIS(-Default-)
Plot type: Mesh Quality1



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Mechanical Drive Pulley Support

Pulley Support-PULLEY SUPPORT STATIC ANALYSIS-Quality-Quality1

Resultant Forces

Reaction forces

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N	444.799	88.958	0.00399351	453.608

Reaction Moments

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N.m	0	0	0	0

Free body forces

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N	0.0906204	0.0677994	-0.0343082	0.118262

Free body moments

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N.m	0	0	0	1e-33

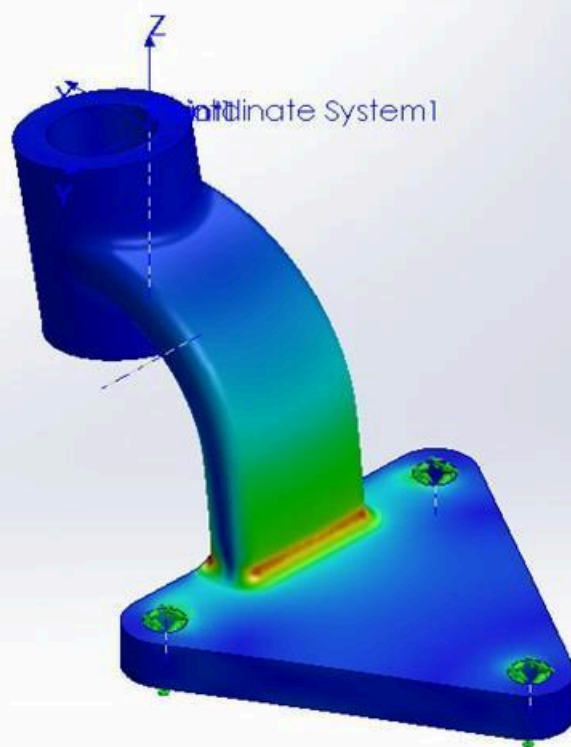
Study Results

Name	Type	Min	Max
Stress1	VON: von Mises Stress	1.458e+01psi Node: 21746	1.919e+04psi Node: 3082



Mechanical Drive Pulley Support

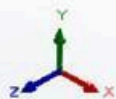
Model name: Pulley Support
Study name: PULLEY SUPPORT STATIC ANALYSIS(-Default-)
Plot type: Static nodal stress Stress1
Deformation scale: 51.6595



von Mises (psi)

1.919e+04
1.727e+04
1.535e+04
1.344e+04
1.152e+04
9.601e+03
7.683e+03
5.766e+03
3.849e+03
1.932e+03
1.458e+01

→ Yield strength: 3.499e+04



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Pulley Support-PULLEY SUPPORT STATIC ANALYSIS-Stress-Stress1



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Analyzed with SOLIDWORKS Simulation

Simulation of Pulley Support

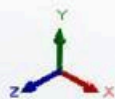
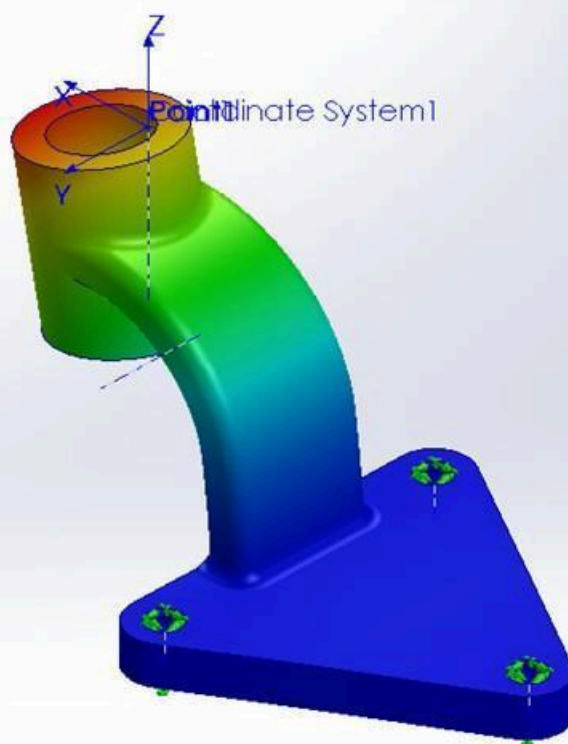
Mechanical Drive Pulley Support

Name	Type	Min	Max
Displacement1	URES: Resultant Displacement	0.000e+00in Node: 19	9.347e-03in Node: 2346



Mechanical Drive Pulley Support

Model name: Pulley Support
Study name: PULLEY SUPPORT STATIC ANALYSIS(-Default-)
Plot type: Static displacement Displacement1
Deformation scale: 51.6595



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Pulley Support-PULLEY SUPPORT STATIC ANALYSIS-Displacement-Displacement1



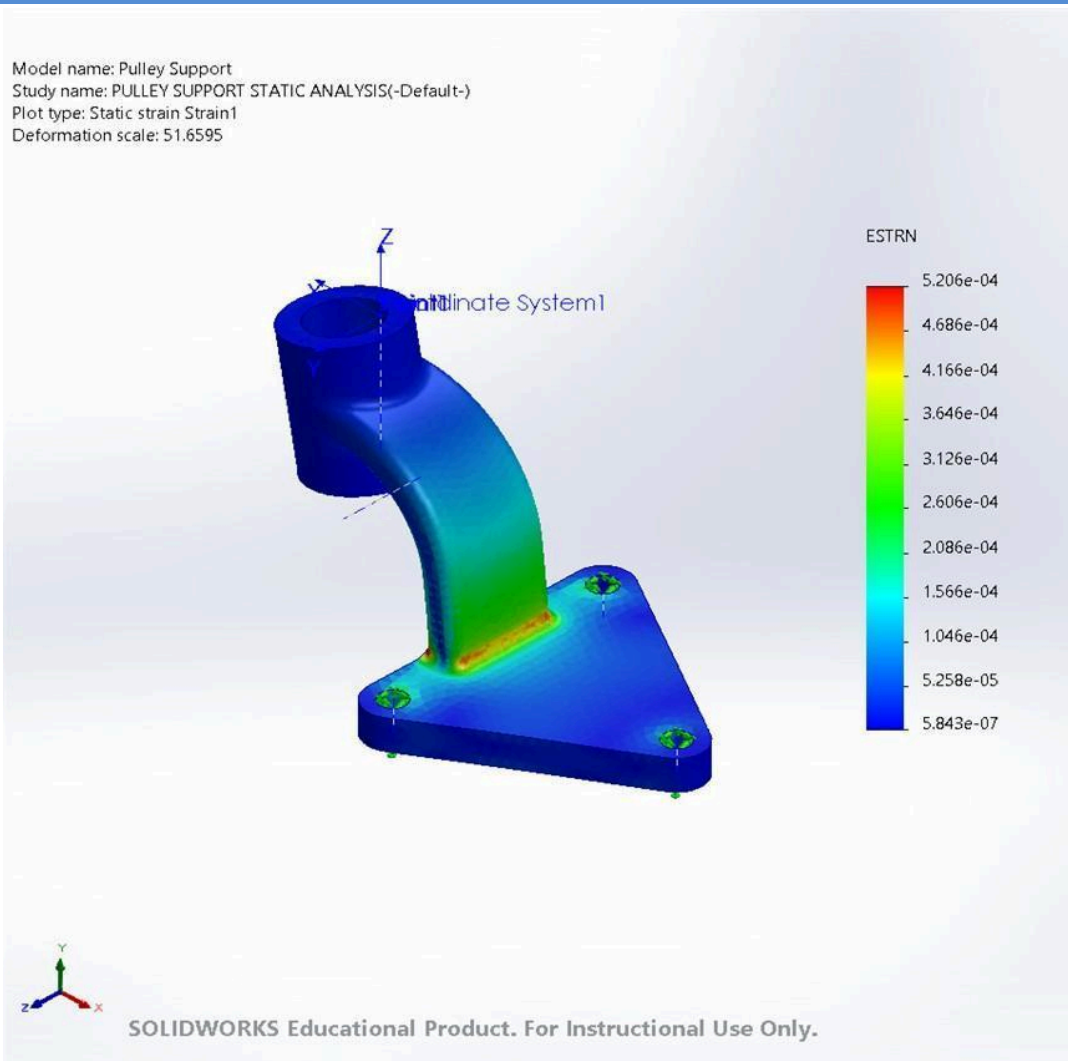
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Analyzed with SOLIDWORKS Simulation

Simulation of Pulley Support

Mechanical Drive Pulley Support

Name	Type	Min	Max
Strain1	ESTRN: Equivalent Strain	5.843e-07 Element: 22989	5.206e-04 Element: 17276



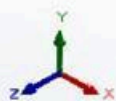
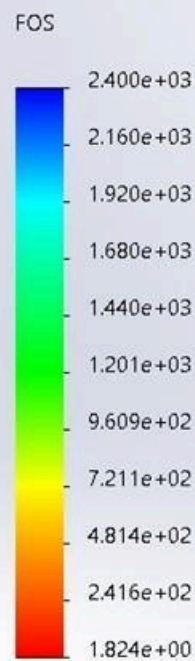
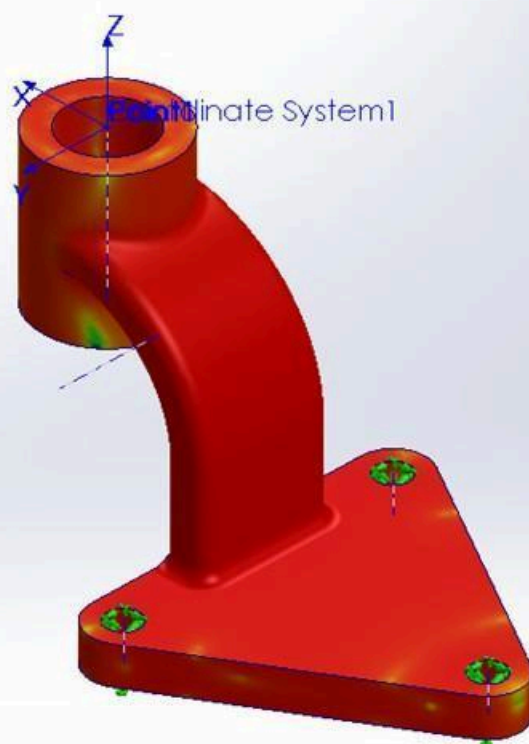
Pulley Support-PULLEY SUPPORT STATIC ANALYSIS-Strain-Strain1

Name	Type	Min	Max
Factor of Safety1	Max von Mises Stress	1.824e+00 Node: 3082	2.400e+03 Node: 21746



Mechanical Drive Pulley Support

Model name: Pulley Support
Study name: PULLEY SUPPORT STATIC ANALYSIS(-Default-)
Plot type: Factor of Safety Factor of Safety1
Criterion : Max von Mises Stress
Factor of safety distribution: Min FOS = 1.8



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Pulley Support-PULLEY SUPPORT STATIC ANALYSIS-Factor of Safety-Factor of Safety1



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Simulation of Pulley Support

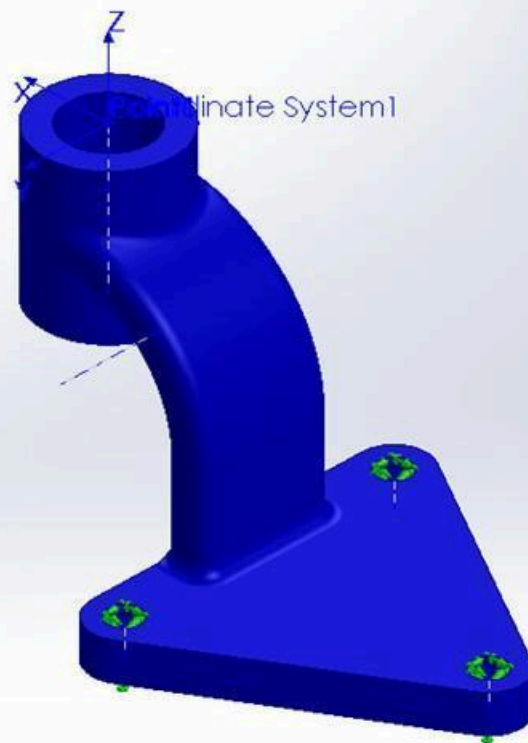
Mechanical Drive Pulley Support

Name	Type	Min	Max
Factor of Safety2	Max von Mises Stress	1.824e+00 Node: 3082	2.400e+03 Node: 21746



Mechanical Drive Pulley Support

Model name: Pulley Support
Study name: PULLEY SUPPORT STATIC ANALYSIS(-Default-)
Plot type: Factor of Safety Factor of Safety2
Criterion : Max von Mises Stress
Red < FOS = 1 < Blue



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Pulley Support-PULLEY SUPPORT STATIC ANALYSIS-Factor of Safety-Factor of Safety2

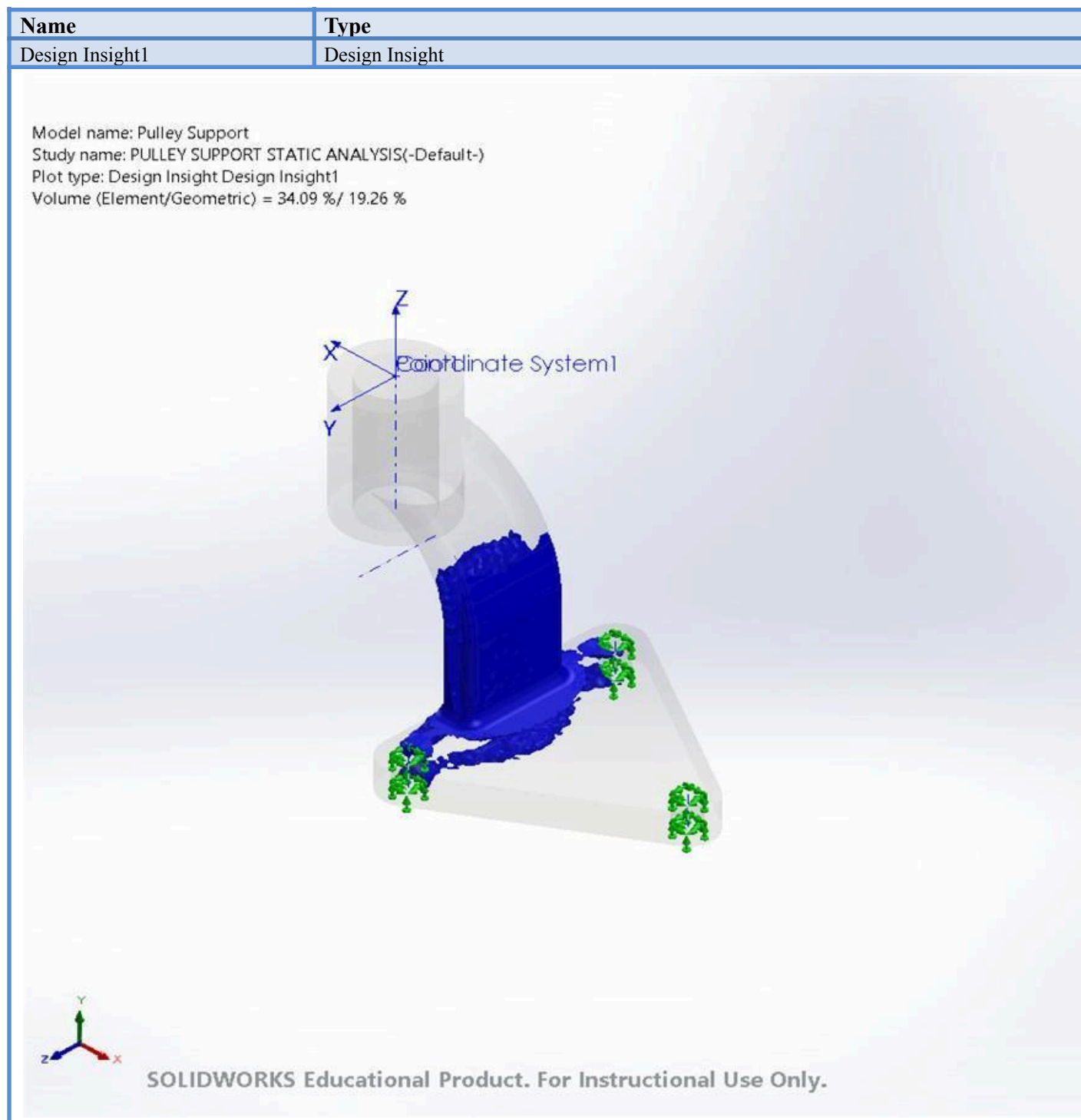


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Analyzed with SOLIDWORKS Simulation

Simulation of Pulley Support

Mechanical Drive Pulley Support



Mechanical Drive Pulley Support

Pulley Support-PULLEY SUPPORT STATIC ANALYSIS-Design Insight-Design Insight1

Conclusion

Stress and Structural Safety:

- Under the applied 100 lbf bearing load and 20 lbf vertical weight, the maximum von Mises stress reached 19,190 psi.
- This maximum stress occurs at Node 3082, located right along the base fillet where the curved neck connects to the flat mounting plate.
- Because the maximum stress of 19,190 psi is well below the material yield strength of 34,994 psi, the bracket will not yield or deform permanently.

Factor of Safety (FOS):

- The minimum Factor of Safety for the part is 1.824 at Node 3082.
- The Factor of Safety plot shows the entire bracket in solid blue, confirming that zero areas fall below the failure limit of $FOS = 1.0$.
- The design is safe for the specified static working loads.

Displacement and Deflection:

- The maximum displacement is 0.0093 inches, occurring at Node 2346 on the top face of the upper cylinder.
- This small deflection is caused by the bending moment created by the horizontal bearing load along the cantilevered neck. The deflection is minimal and will not interfere with pulley alignment.
- The maximum equivalent strain is $5.21e-04$ at Element 17276, matching the high-stress fillet area.

Design Insight and Optimization:

- The Design Insight plot reveals that the load path runs directly through the curved neck and the center of the baseplate.
- More than 65% of the bracket material carries little to no load, specifically the outer edges of the base and the sides of the cylinder.
- For future production runs, material can be removed or pocketed from these low-stress regions to reduce part weight and manufacturing cost without sacrificing strength.

